STANDATA variance 23-BCV-010R1

Building

Engineer's authentication for welding procedures

Date Issued: April 2025

Purpose

To provide an alternative solution to the National Building Code – 2023 Alberta Edition (NBC(AE)) requirement for certification of fabricators to CSA W47.1 – "Certification of companies for fusion welding of steel", for the fabrication of skids for mounting mechanical equipment/systems or fabrication of proprietary structural components.

Discussion

Buildings and their steel structures are required to comply with CSA S16:19 "Design of steel structures" through Sentence 4.3.4.1.(1) of the NBC(AE). The requirement to comply with CSA W47.1 "Certification of companies for fusion welding of steel" stems from the reference to it in Clause 24.3 in CSA S16, which states fabricators and erectors responsible for welding structures fabricated or erected under CSA S16 shall be certified by the Canadian Welding Bureau (CWB) to the requirements of CSA W47.1 (Division 1 or 2), CSA W55.3, or both, as applicable.

Fabricators outside of Canada have limited access to CWB certified companies, and thus, it is much more difficult for them to be deemed to comply with CSA W47.1. The standard commonly adopted by the international community of fabricators involved in the fabrication and welding of structural steel is AWS D1.1 "Structural Welding Code-Steel." When compared to CSA W47.1, AWS D1.1 does not contain all the criteria that CSA W47.1 speaks to; therefore, it is necessary for this variance to provide equivalent measures in the form of engineering quality controls in order for the outcome to be approximately equivalent or greater in safety. The inclusion of engineering quality controls together with the requirements of AWS D1.1 can be viewed as an acceptable solution to the requirements of CSA W47.1.

Similar to CSA W59 "Welded Steel Construction" (metal arc welding), AWS D1.1 "Structural Welding Code–Steel" influence extends globally, and it is common to find countries that adopt it as a reference or modify it according to local regulations. The Canadian equivalent standard to AWS D1.1 is CSA W59 "Welded Steel Construction" (metal arc welding). The two standards are quite similar with respect to welding requirements; however, another standard, CSA W47.1 "Certification of companies for fusion welding of steel", provides certification requirements for the welding companies. There is no equivalent standard to complement CSA W47.1. Companies following AWS D1.1 requirements for welding activities do so voluntarily without third party certification for their welding procedures and rely on the company's own quality management plan.

A variance involving the practice of engineering is also a matter for the Association of Professional Engineers and Geoscientists of Alberta (APEGA) to take into consideration, as the practice of engineering is a self-regulated profession governed by APEGA. The references to APEGA's authentication standard as a condition of this variance is a substantive basis for this variance providing approximately equivalent or greater safety performance.

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Code References

Division A, Article 1.4.1.2.

1.4.1.2. Defined Terms
1) The words and terms in italics in this Code shall have the following meanings:

Registered engineering professional means an individual who is authorized to engage in the practice of engineering under the Engineering and Geoscience Professions Act and its Regulations.

Sentence 4.3.4.1.(1)

4.3.4.1. Design Basis for Structural Steel
(1) *Buildings* and their structural members made of structural steel shall conform to CSA S16, “Design of Steel Structures.” (See Note A-4.3.4.1.(1).)

Division C, Sentence 2.2.2.4.(1)

2.2.2.4 Fabrication and Erection of Steel
1) If requested by the authority having jurisdiction, the owner shall submit evidence, before the project begins, that all fabricators and erectors of welded construction for buildings designed in accordance with Part 4 of Division B are certified by the Canadian Welding Bureau.

Application

This variance applies to fabricators not certified to CSA W47.1 for the fabrication of skids for the mounting of mechanical equipment/systems or fabrication of proprietary structural components.

Variance

This variance provides approximately equivalent or greater safety performance with respect to persons and property as provided for by the *Safety Codes Act* and the NBC(AE).

The following alternative measures are deemed as compliant with the NBC(AE) for fabricators not certified to the CSA W47.1:

1. All documentation as determined by the registered engineering professional must be authenticated in accordance with APEGA’s authentication standards and must be submitted to the authority having jurisdiction.

2. The requirements of AWS D1.1 relevant to welding procedures are complied with.

3. A registered engineer professional ensures the measures for equivalency in column #4 of Table “A” are achieved.

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Table "A"
Comparison Table Reference

Comparison	Canadian Standard CSA W47.1 As referenced in CSA S16:19	International Standard AWS D1.1:2000	Measures for Equivalency
SCOPE	This Standard provides the requirements for the certification of companies engaged in the fusion welding of steel and the qualification of their personnel and welding procedures.	This code contains the requirements for fabricating and erecting welded steel structures. When this code is stipulated in contract documents, conformance with all provisions of the code must be required, except for those provisions the Engineer or contract documents specifically modifies or exempts	Owner's due diligence by verifying the fabricator has a quality (including welding) program in place. This can be achieved during the review process of fabricator quality/welding documents by registered engineering professional and owner assigned inspector during the fabrication activities
	Is administered by CWB, which is mandated to certify fabricators and monitor their conformance to Standards requirement.	AWS D1.1 is a voluntary conformance standard; AWS does not monitor and enforce the conformance.	The registered engineering professional verifies the competency of fabricator.
QUALIFIED WELDING ENGINEERS	Companies must employ/retain a CWB qualified welding engineer (Division 1 and 2 only).	No requirement.	Owner or representative captures the requirement of engaging with a registered engineering professional either as a fabricator employee or third-party consultant in the procurement contract/technical documentation.
QUALIFIED WELDING SUPERVISORS	All certified companies are required to employ at least one CWB qualified Welding Supervisor.	No requirement.	Owner or representative captures the requirement of engaging with a welding supervisor either as a fabricator employee or third-party consultant in the procurement contract/technical documentation.
QUALIFIED WELDERS	Qualification Process: As acceptable to CWB.	Qualification Process: As acceptable to Engineer.	As acceptable to a registered engineering professional and is verified by the owner assigned inspector during the shop inspection process.

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Comparison	Canadian Standard CSA W47.1 As referenced in CSA S16:19	International Standard AWS D1.1:2000	Measures for Equivalency
	Qualification Responsibility: Fabricator/manufacturer; however, requires witnessing by CWB. Union, Test Centre, Education Institutions require CWB accreditation and witnessing by CWB.	Qualification Responsibility: Fabricator/manufacturer does not require witnessing by AWS. Union, Test Centre, Education Institutions require AWS accreditation however witnessing by AWS is not required.	Owner assigned inspector must review and verify welders' qualifications during the shop inspection process.
	Period of effectiveness: Two years.	Period of effectiveness: Indefinite.	Review and verification of a welder continuity log or any other shop-maintained document by owner's inspector.
QUALIFIED WELDING PROCEDURES	Qualification Process: Acceptable to CWB.	Qualification Process: Acceptable to Engineer.	Acceptable to the registered engineering professional.
	Qualification Responsibility: Fabricator/Manufacturer requires witnessing and approval by CWB.	Qualification Responsibility: Fabricator/Manufacturer-witnessed by accredited labs or recognized certification bodies acceptable to Engineer.	Acceptable to the registered engineering professional.
	Witnessing of testing: Only by CWB or authorized rep, CWB is the independent administrator.	Witnessing of testing: No independent administrator.	Owner assigned inspector must review and verify the applicability during the shop inspection.
	Approval of procedures: CWB is the independent administrator, approval only by CWB.	Approval of procedures: Not applicable, no independent administrator required.	Approval by the registered engineering professional.

This VARIANCE is applicable throughout the Province of Alberta.

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Unless stated otherwise, all Code references in this STANDATA are to Division B of the National Building Code - 2023 Alberta Edition

Issued by the Provincial Building Administrator

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